



EASTERN VISAYAS STATE UNIVERSITY – ORMOC CAMPUS
COMPUTER STUDIES DEPARTMENT

**TRIMINDER: DEVELOPMENT OF A GAMIFIED SCREEN TIME MANAGEMENT
AND WELLNESS MONITORING APPLICATION FOR PROMOTING HEALTHY
DIGITAL HABITS AMONG EVSU ORMOC STUDENTS**

A Capstone Project Proposal
Presented to the Faculty of the Computer Studies Department
In Partial Fulfillment of the Requirements
for the Degree
Bachelor of Science in Information Technology

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APPROVAL SHEET

This Capstone Project Proposal entitled:

**“TriMinder: Development of a Gamified Screen Time Management and Wellness
Monitoring Application for Promoting Healthy Digital Habits among EVSU Ormoc
Students”**

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ABSTRACT

This Capstone Proposal presents the development of TriMinder, a development of a gamified screen time management and wellness monitoring application for promoting healthy digital habits among Eastern Visayas State University - Ormoc Campus Students. The project addresses the shortcomings of existing digital wellness applications, which often rely on static and dismissible reminders that fail to encourage behavioral change. BreadBuddy aims to introduce a more interactive and persuasive approach by incorporating automated screen break enforcement and gamified incentives such as progress tracking and rewards. The system is designed for Android smartphones and is tailored to meet the needs of individual users. Utilizing the Agile development methodology, the application will be built iteratively with continuous user feedback, ensuring a user-centered design. Core features include screen usage monitoring and non-restrictive alerts. A trial deployment will assess user compliance, screen time reduction, and satisfaction. BreakBuddy is expected to enhance digital wellness through behaviorally informed engagement strategies. Its significance extends to users, offering a practical tool for fostering conscious screen use and reducing technology-related strain. Additionally, it contributes to the field of information technology by applying gamification and behavioral science principles in mobile app development, serving as a foundational study for future research in digital wellness solutions.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

As digital technology becomes an integral part of everyday life, mobile devices have become central to students' academic, social, and personal lives. According to Datareportal (2024), Filipino internet users spend an average of 8 hours and 52 minutes online daily—among the highest in the world. Excessive screen time, particularly among college students, is associated with adverse effects including reduced academic performance, digital fatigue, poor posture, disrupted sleep, and diminished mental well-being (World Health Organization [WHO], 2023). Disruptions in sleep and concentration due to prolonged screen exposure have also been widely documented in health literature.

At Eastern Visayas State University (EVSU), Ormoc Campus, students from various departments are increasingly reliant on smartphones for both academic and personal purposes. However, many lack self-discipline in managing their screen time, affecting productivity and overall wellness. There is currently no unified platform for students or administrators to monitor or encourage responsible device usage. The campus community has yet to implement a gamified approach that combines wellness advocacy, digital discipline, and peer motivation to improve student habits.

Previous efforts and tools that promote screen monitoring lack community engagement, gamification, or incentive-based systems. These gaps present a significant opportunity to implement a solution tailored for the EVSU context—specifically, a gamified mobile application that empowers students to track, reflect on, and improve their screen usage behavior.

According to our survey conducted among more than 200 EVSU Ormoc students, the most frequently used applications include TikTok, Facebook, YouTube, Messenger, and Instagram—indicating that excessive screen time is often driven by entertainment and social media use. This reinforces the urgency of managing digital habits through a structured, student-centered system.

This capstone project proposes the development of a mobile-based ~~gamified screen time~~ management and wellness monitoring application—available for both Android and iOS devices—for EVSU Ormoc College students. The project also includes a ~~web-based admin~~ platform where the Supreme Student Government (SSG) serves as super admins, granting access to department-based sub-admins. ~~This system enables visibility into the ranking of digitally disciplined students. It promotes peer-to-peer encouragement via badges, likes, and profiles—all contributing to building a campus culture of healthy digital habits.~~

Additionally, the system allows students to connect with their peers by adding friends within the app. Once connected, students can view their friends' screen time performance through a dedicated "Friends ranking" and can visit their friends' profiles to give or remove likes as a form of digital support. Users can also *gain likes from others*, which adds to their motivation and boosts their profile performance. The same interaction is available within their department ranking, enabling users to show appreciation for top-performing classmates. SSG super admins and department-based sub-admins also can like or unlike user profiles, providing a non-intrusive way to recognize outstanding digital wellness behavior across the platform.

✗ Give only the generalized purpose
↳ everything [details] can be outlined in the objectives

1.2 Problem Statement

College students at EVSU Ormoc Campus increasingly struggle with excessive and unregulated screen time. This trend negatively impacts their academic focus, sleep routines, physical activity, and mental wellness. Existing tools such as Digital Wellbeing offer basic monitoring features but lack motivation mechanisms, gamification, or community engagement aspects that promote consistent usage and behavior change.

There is no centralized platform at EVSU that enables student leaders to monitor trends in screen usage, celebrate productive digital behavior, or distribute wellness information. Moreover, student organizations have limited tools to promote digital discipline within their respective departments. Likewise, students lack peer-to-peer engagement features that can foster encouragement, recognition, and healthy competition.

Without an innovative, engaging, and data-driven system, students are left without structured support in developing healthier screen habits, and administrators are unable to identify patterns of concern or top-performing individuals in terms of digital wellness. Thus, a gamified and campus-centric solution is urgently needed to address these gaps in digital behavior management.

1.3 General Objectives

To develop a gamified screen time management and wellness monitoring application for promoting healthy digital habits among EVSU Ormoc College students.

1.4 Specific Objectives

- To design and develop a mobile application (for Android) that allows students to monitor app usage, set goals, and receive wellness suggestions (e.g., stretching, meditation).

In the intro [background] the discussion includes Android & iOS

Can be rewritten in a more concise form
→ get to the specifics directly

- To implement gamification features such as XP points, digital discipline badges, and daily performance rewards based on screen time.
- To build a ranking system viewable by students, ~~SSG super admins~~, and department-based sub-admins to highlight top-performing students in terms of digital discipline. *Should never be for SSG → SASO*
Guidance Of
- To allow users to add friends and access a "Friends Ranking" where they can view and interact with their friends' digital performance.
- To allow students to like or unlike their friends' profiles and interact with the department ranking by liking or unliking top-performing users—and for users to gain likes from others.
- To allow SSG and sub-admins to like or unlike student profiles as a form of recognition and encouragement.
- To develop a web-based platform for the SSG that facilitates admin access control, student ranking management, and viewing of student wellness profiles.
- To allow department-based organizations to access filtered ranking data of students within their specific department and promote healthy screen habits through public recognition.
- To conduct usability testing and evaluation based on ISO/IEC 25010 software quality standards.

Why?

1.5 Scope Limitations

Scope

This project involved the development of:

- A mobile application (for Android) for students to:
 - Track screen usage and receive notifications for healthy breaks (e.g., “stand up and stretch”).
 - Collect gamified rewards such as XP points and badges for disciplined behavior.
 - Add other users as friends and view their screen time performance via a "Friends ranking."
 - Like or unlike profiles of friends or department peers as a way of peer encouragement.
 - Gain likes from friends or peers to boost their public profile and motivation.
 - Read the terms and conditions when registering or logging in to the application.
- A web platform for the SSG (super admins) to:
 - View daily ranking of top-performing students across all departments.
 - View individual student profiles with badges, screen usage summaries, and community engagement stats (e.g., number of likes).
 - Add department organizations (sub-admins) and manage access privileges.
 - Users can like or unlike a student profile as part of recognition efforts.
- Department-based sub-admins can:
 - View ranking filtered by department (e.g., only Computer Studies students).
 - Promote top students' performance through their organization initiatives.

Conform With the Possible Facts
in Problem & Objectives

- Have the option 'like' or 'undo like' on student profiles within their department.
- The public homepage of the admin website will allow students to:
 - Instructions for installing the Android app.
 - Download the Android app.

Limitations

The app is limited to Android devices only; iOS is not supported. The mobile app requires user permissions to monitor screen time and access notifications. The ranking refreshes daily and cannot be altered by admins or users. Users can only like or unlike a specific user once per day to avoid spamming. Usability testing will be limited to a selected number of students from different departments due to time and resource constraints. An Internet connection is required for syncing rewards, badges, friend data, and ranking information.

1.6 Significance of the Study

This capstone project is primarily beneficial to EVSU Ormoc College students, who struggle to manage their screen time across academic and personal activities. By offering a mobile-based system with gamified features, students are empowered to track usage, receive health-conscious break prompts, and compete for ranking, which can encourage discipline and healthier digital habits. The ability to add friends, like profiles, and view department-specific ranking promotes community support and student-driven motivation. The likes received from peers serve as social recognition, increasing user engagement and accountability.

According to our survey, students and SSG representatives emphasized the value of features such as customizable break reminders, battery-efficient design, and friend-based

Refine it more, make the proposal work without making the system too easy. This is not a regular Project.

**NOTE: SSG has no control over the students.
LET S&SO, Guidance Office handle it.**

7

rankings—highlighting the importance of user-centered functionality and peer interaction in promoting consistent usage and motivation.

The Supreme Student Government (SSG) and department-based organizations will also benefit by gaining access to reliable, real-time data on student screen time behavior. This can be used for identifying role model students, recognizing improvement trends, and creating campaigns to promote balanced digital habits campus-wide. Their ability to like or unlike students' performance profiles also adds a layer of engagement and recognition within student governance, encouraging a culture of positive digital discipline.

Secondary beneficiaries include faculty, guidance counselors, and researchers who may use the data trends generated by the platform to understand student well-being better. Parents may also indirectly benefit as their children adopt healthier digital routines, reducing mental strain and improving academic outcomes. Developers and IT educators can use this project as a model for integrating wellness with mobile gamification, providing practical value in both instructional and applied settings.

Technologically, the system promotes digital transformation within educational institutions by replacing passive screen time alerts with an interactive, gamified, and socially connected platform. The integration of modern features such as cross-platform availability, real-time updates, and peer-based interaction supports a shift toward student-centered digital wellness solutions.

Furthermore, this project aligns with institutional modernization and the broader goals of inclusive access to digital health solutions, supporting the Commission on Higher Education's (CHED) advocacy on digital wellness and productivity in higher education institutions.

fbcs

Refine

1.7 Definition of Terms

Gamification – The application of game-design elements (e.g., points, badges, ranking) to non-game contexts such as wellness and education.

Screen Time – The total amount of time a user spends interacting with a digital screen, typically measured in hours per day.

Ranking – A ranked list of users based on their performance metrics such as screen time discipline, XP points, or rewards.

Sub-admin – A user account with partial administrative privileges assigned to department-based organizations.

Digital Wellbeing – The state of an individual's physical and mental health concerning their digital device usage.

Friends ranking – A personalized ranking that displays only the performance of users added as friends within the application.

Like/Unlike Feature – A social feedback mechanism allowing users to express support or recognition for another user's profile performance.

ISO/IEC 25010 – An international standard for evaluating the quality of software systems based on characteristics such as usability, efficiency, and reliability.

Gained Likes – The total number of likes a user receives from others in the app, reflecting peer recognition of their digital discipline.

Define

Break Notification – An automated reminder triggered by the app prompting the user to take a health-conscious break, such as stretching, resting eyes, or walking, based on screen usage duration.

XP (Experience Points) – A gamification metric awarded to users for practicing digital discipline (e.g., reduced screen time), used to track progress and unlock rewards or badges within the system.

Digital Discipline Badge – A virtual reward granted to users who consistently meet screen time goals or demonstrate positive digital wellness behaviors.

Department-Based Ranking – A leaderboard that displays the screen time performance of students filtered by academic department, allowing localized recognition and comparison.

Undo Like – A feature that allows users to remove a previously given like on another user's profile, limited to once per day, to ensure fair and non-spammy engagement.

CHAPTER 2: REVIEW OF RELATED LITERATURE

2.1 Introduction

This chapter presents a thematic review of literature and empirical studies relevant to gamified mobile applications, screen time management, wellness interventions, and peer-supported digital discipline systems. The reviewed themes include: (1) Screen time behavior and consequences; (2) Gamification in mobile wellness and education; (3) Peer interaction and social motivation; (4) Existing screen time apps and their limitations. The chapter concludes by synthesizing key findings and identifying research gaps that justify the proposed system.

2.2 Thematic Review of Literature

Screen Time Behavior and Consequences

Liew and Aung (2021) conducted a study showing that excessive smartphone usage among university students significantly disrupts sleep quality, leading to heightened stress, fatigue, and reduced academic performance (IJABIM). These outcomes are consistent with WHO (2023) recommendations on sedentary behavior and mental well-being.

Gamification in Mobile Learning and Wellness

Gamification in educational and wellness mobile applications has shown a significant impact in improving engagement and learning outcomes. A meta-analysis conducted across Asian educational settings revealed a substantial effect size ($g \approx 1.3$), underscoring the consistent positive influence of gamification on student performance and motivation (ERIC). Similarly, Lampropoulos and Sidiropoulos (2024) presented a longitudinal study that demonstrated how

gamified learning environments significantly improved student success rates by 39%, excellence rate by 130%, and retention by 42% compared to traditional and non-gamified online setups. These results confirm the capacity of gamification to foster sustained academic motivation. Furthermore, Pechenkina et al. (2017) observed that a gamified quiz application that incorporated badges and ranking increased university students' retention and academic performance. These studies confirm the effectiveness of gamified systems in enhancing cognitive engagement and motivation, particularly in higher education contexts.

Social Engagement and Peer Influence

Social interaction and peer-based gamification strategies have been identified as powerful motivators in behavior change. In a study by Shameli et al. (2017), walking challenges implemented through a mobile app led to an average 23% increase in daily physical activity, attributed to the presence of friendly competition. This finding illustrates that incorporating ranking and peer challenges within digital applications can create a sense of accountability and shared progress. Moreover, a study published in the *Journal of Information Systems Engineering and Management* (2025) found that student engagement increased by up to 66.7% in gamified systems that enabled interactive components such as chatbots and peer progress visibility. These results emphasize the importance of social connectedness in sustaining user engagement and improving outcomes in wellness and educational platforms.

Potential Downsides and Motivation Decay

While gamification is widely celebrated for its benefits, researchers have also noted its potential drawbacks when implemented without proper design considerations. Almeida et al. (2023) conducted a systematic review. They highlighted that poorly implemented

gamification—particularly systems relying heavily on ranking and badges—can lead to reduced intrinsic motivation and encourage users to game the system rather than adopt genuine behavioral changes. Over time, the novelty of rewards may wear off, leading to disengagement. These findings emphasize the need for carefully balanced gamified systems that maintain user interest and avoid over-reliance on competitive mechanics.

2.3 Comparative Analysis of Screen Time Management Apps

Table 2.3.1 Comparison of Existing Screen Time Management Applications

Application	Platform	Key Features	Strengths	Limitation
Google Digital Wellbeing	Android	App usage timer, wind-down mode, dashboard	Built-in with Android simple UI	No gamification, no social features
StayFree	Android	Usage tracker, overuse alert, app locking charts	Lightweight and simple	No community-based features
Forest	Android/iOS	Stay focused by planting virtual trees, a rewards-based system	Unique visual gamification promotes focus	Limited scope for wellness tracking
RescueTime	Android/iOS	Detailed activity reports, goal setting, and productivity scoring	Advanced analytics, focus-oriented	Lacks gamification and peer recognition
Space	Android/iOS	Goal setting, screen time reduction coach, insights	Personalized insights	Subscription-based, no social sharing features

YourHour	Android	Time tracking, daily reports, addiction level scoring	Personalized addiction scoring	Too many ads, some features are premium only
Flipd	Android/iOS	Full phone lockouts, wellness content, focus sessions	Strict mode encourages discipline	Invasive for some users, limited social interactivity
ScreenZen	Android/iOS	Self-reflection prompts, mindfulness break, time limits	Combines mental health with screen tracking	Minimal gamification, limited data visualization
LockStock	Android	Rewards for locking the phone during class	Incentivizes offline time in campus contexts	Only works within defined geofences

While all the apps above offer some form of screen time monitoring, few include gamified features, and fewer still provide peer-to-peer interactions such as liking profiles or ranking within campus communities. The proposed BreakBuddy application addresses these limitations by integrating social recognition, gamification, and personalized ranking within a school context.

2.4 Theoretical and Conceptual Framework

Theoretical Foundation

Self-Determination Theory (SDT) underpins the design of the gamified app. It holds that intrinsic motivation is fostered when users feel autonomy (choosing goals), competence (earning XP and badges), and relatedness (connecting with peers via friend ranking) (Pechenkina et al., 2017).

Conceptual Framework

This capstone integrates the SDLC (Software Development Life Cycle) with SDT to model system flow:

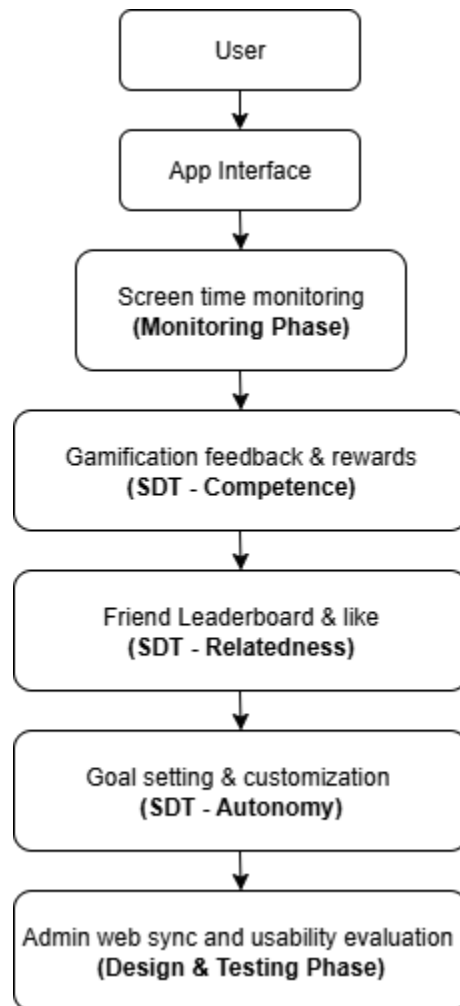


Figure 2.4.1 Conceptual Framework of TriMinder Based on SDLC and Self-Determination Theory (SDT)

2.5 Synthesis and Research Gap

The literature affirms that gamification can significantly elevate engagement, performance, and behavior change. However, several key gaps are evident:

- Existing apps rarely merge screen time tracking, gamification, and peer recognition within a university context.
- There is limited research on campus-driven, Android-only mobile wellness systems.
- The risk of demotivation due to poorly designed ranking or novelty wear-off has not been addressed in most systems.

This capstone project introduces a holistic, gamified Android application tailored for EVSU Ormoc students—equipping users with both quantitative feedback and peer-driven encouragement. At the same time, administrators oversee wellness data and recognize digital discipline, filling a substantive gap in Filipino higher education digital wellness solutions.

CHAPTER 3: METHODOLOGY

3.1 Research Design

This study employs a developmental research design using a mixed-methods approach that integrates both qualitative and quantitative data gathering. The mixed approach is suitable for addressing the practical needs of system development, usability testing, and user-centered design. It enables the collection of rich, contextual data on user behavior and preferences while allowing statistical validation of trends and usability outcomes. The study is primarily developmental, as it focuses on the iterative creation, testing, and refinement of the "TriMinder" mobile and web-based application for screen time management and wellness monitoring.

This design is aligned with the project's aim of building a gamified digital wellness platform specifically tailored to the needs of Eastern Visayas State University (EVSU) Ormoc students. Through user feedback loops and progressive system sprints under the Agile methodology, the design will ensure continuous improvement based on both empirical insights and user engagement data.

3.2 Data Collection Methods

Data for this study will be collected using structured survey questionnaires. These surveys will be distributed to the following respondents:

- EVSU Ormoc College students from any year level (general users)
- Student organizations per department (serving as sub-admins)
- Supreme Student Government (SSG) officers (super admin perspective)

A minimum of 100 student participants will be targeted using **purposive sampling**, focusing on those who are active smartphone users. Each department will have one officer to represent sub-admins. One or more representatives from the SSG (e.g., president or digital wellness officer) will be selected as key informants.

Among students surveyed, app usage patterns and preferred features like dark mode, wellness notifications, and non-intrusive ranking interactions were prioritized.

Survey distribution and consent:

- Student participants will receive an anonymous consent-based questionnaire either online or via printed forms. Participation is voluntary, and no identifiable personal data will be collected.
- Student organization officers and SSG respondents will be asked to sign a formal letter of informed consent acknowledging their participation and understanding of the project.

The data collected will cover:

- Digital usage behavior (daily screen time, types of apps used)
- Awareness and perceptions about digital wellness
- Preferences for gamification features (XP, badges, ranking, friend system)
- Perceived usefulness and interest in a wellness tracking system

Ethical Considerations

All participants will be informed about the research objectives and assured that their participation is entirely voluntary. They will have the right to withdraw at any time without consequence. Informed consent will be obtained from all participants, and in the case of minors,

parental or guardian consent will be secured before their participation. The study will not collect any personally identifiable information, and all responses will be treated with strict confidentiality. Data will be stored securely and used solely for academic purposes, according to established ethical research standards.

3.3 Data Analysis Techniques

Quantitative data from student surveys will be processed using descriptive statistics, including frequencies, percentages, means, and standard deviation to understand digital behavior and design preferences. These statistics will help prioritize features such as which rewards, alerts, or ranking interactions are most engaging.

The qualitative responses (especially from the comment section) were categorized thematically (e.g., concerns about intrusiveness, ranking pressure, cross-device tracking, battery optimization, etc.).

Qualitative data from student organization and SSG respondents will be thematically coded into categories such as:

- Recognition practices for digital wellness
- Desired admin features and ranking controls
- Feedback mechanisms and motivational strategies

The combination of quantitative and qualitative analysis will guide system design decisions and provide a foundation for usability testing, validation of UI/UX elements, and enhancement of gamified engagement mechanisms.

3.4 System/Software Design

The **Agile Software Development Life Cycle (SDLC)** model will be used in this project. Agile is selected for its iterative nature, ability to accommodate ongoing feedback, and suitability for user-centered system design. It is particularly beneficial in academic projects that involve multiple stakeholders and require progressive refinement.

The Agile phases to be followed include:

- **Planning:** Define features based on research objectives and survey feedback.
- **Analysis:** Analyze survey data to determine user preferences and system requirements.
- **Design:** Create wireframes and interface mockups using user requirements.
- **Implementation:** Code mobile and web-based modules through sprint cycles.
- **Testing:** Perform unit and integration testing, and conduct user-based usability evaluations.
- **Deployment:** Prepare the application for pilot release within the EVSU Ormoc campus.
- **Maintenance:** Address feedback, bugs, or enhancements identified during testing.

3.5 Tools and Technologies

The development of TriMinder will utilize a combination of modern frameworks, backend services, and developer tools to ensure a responsive user experience, efficient code management, and scalable architecture.

- **Development Platform and Backend Services:**
 - **Supabase** will be used as the Backend-as-a-Service (BaaS) platform. It provides real-time database features, authentication, and RESTful APIs, making it suitable

for managing user data, authentication, screen activity logs, and gamified rewards. Supabase's open-source PostgreSQL infrastructure ensures reliability and scalability throughout the application's lifecycle (Supabase, n.d.).

- **SQLite** will also be integrated for local storage purposes. As a lightweight, embedded SQL database engine, SQLite is ideal for storing data directly on the user's device, providing offline access to essential data such as cached activity logs and user preferences. Its serverless design and zero-configuration setup make it efficient for mobile and web applications (SQLite, n.d.).

- **Programming Language and Framework:**

- The system will be developed using **Flutter**, a cross-platform open-source UI toolkit developed by Google. Flutter enables the creation of natively compiled applications from a single codebase, ideal for building responsive, interactive, and gamified mobile applications. It supports high-performance rendering and a widget-based architecture well-suited for both children and adult users (Google Developers, n.d.).

- **Code Editor:**

- **Visual Studio Code (VS Code)** will serve as the primary code editor. VS Code offers advanced extension support for Flutter and Dart, built-in debugging tools, Git integration, and code intelligence features, making it an efficient tool for team-based development and testing (Microsoft, n.d.).

- **Design and Modeling Tools:**

- **Draw.io** (also known as diagrams.net) will be used for creating system architecture diagrams, wireframes, data flow diagrams, and Entity-Relationship Diagrams (ERDs). These visual tools aid in both planning and documenting the software development process (Diagrams.net, n.d.).
- **Lucidchart** will be used as a visual diagramming tool to aid in the design and planning of system architecture, database schemas, and process flows. Its collaborative interface allows the development team to visualize relationships and workflows clearly, improving communication and accuracy in system design (Lucidchart, n.d.).
- **Figma** will serve as the primary user interface (UI) and user experience (UX) design tool. It allows the team to design high-fidelity prototypes, wireframes, and mockups in a collaborative environment. Figma's cloud-based platform enables real-time design feedback and efficient iteration across multiple team members (Figma, n.d.).

- **Project Management and Collaboration:**

- **GitHub** will serve as the project's version control and collaboration platform. Through GitHub, developers will manage code repositories, track issues, conduct code reviews, and enable continuous integration workflows, ensuring codebase integrity and project transparency across the team (GitHub Docs, n.d.).

CHAPTER 4: NEEDS ASSESSMENT AND REQUIREMENT ANALYSIS

4.1 Needs Assessment

Current Situation Analysis

Currently, students at Eastern Visayas State University (EVSU), Ormoc Campus, have no unified system to track and manage their screen time. While some rely on built-in tools like Digital Wellbeing, these apps lack motivation features, social engagement, and gamified elements that are proven to enhance user retention and behavior change.

Students commonly attempt manual self-regulation but is often ineffective due to the absence of peer accountability, feedback mechanisms, or structured interventions. Additionally, no administrative tools exist for the Supreme Student Government (SSG) or departmental student organizations to monitor digital behavior trends or promote wellness initiatives.

Several inefficiencies were identified:

- There is no incentive system for students to practice digital discipline.
- Lack of peer-based engagement in digital wellness efforts.
- Fragmented tracking tools that do not include department-based visibility.
- No cross-device tracking, with many students reporting higher screen time on laptops than phones.

Stakeholder Input

A. Quantitative Results from Student Survey

Out of over 200 student respondents from EVSU Ormoc Campus, the majority reported heavy daily screen usage, averaging between 5 and 8 hours per day, with many also admitting to poor focus and time management. Approximately 85% used mobile phones as their primary device, while a smaller percentage used laptops or tablets. Most popular apps used include Facebook, TikTok, YouTube, and Messenger, indicating intense exposure to time-consuming digital platforms.

Regarding digital wellness solutions, students overwhelmingly favored features like break notifications (91%), XP-based rewards (84%), and daily screen time goals (87%). Interestingly, a substantial number also supported public elements such as leaderboards (73%) and friends' visibility (70%), validating the inclusion of gamified and social accountability features in TriMinder.

Q1. Average Smartphone Usage per Day

1. On average, how many hours a day do you use your smartphone? (e.g., 5 Hours)
211 responses

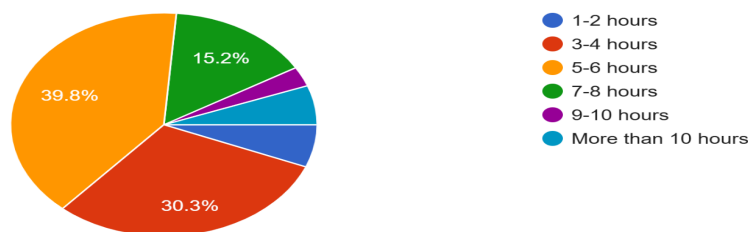


Figure 4.1.1 Average Smartphone Usage per Day

The data presented in Figure 4.1 highlights that the majority of student respondents—particularly those from EVSU Ormoc—reported using their smartphones between 5 and 6 hours daily. This prolonged screen time suggests a high level of dependency on digital devices, particularly for social, academic, and entertainment purposes. Such a pattern of usage supports the need for an intervention tool like TriMinder, which promotes screen time awareness and healthy digital habits through gamified features.

Q2. Primary Devices Used by Students

2. What type of device do you primarily use?
211 responses

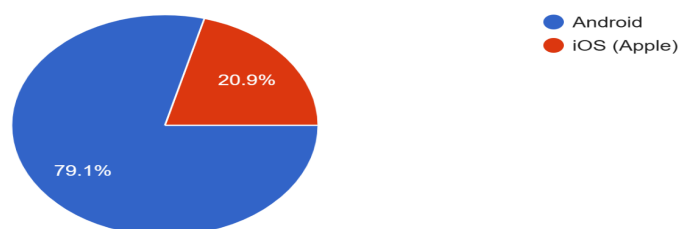


Figure 4.1.2 *Primary Devices Used by Students*

Figure 4.2 illustrates that 79.1% of respondents primarily use Android phones, followed by 20.0% who use iOS devices. This confirms that an Android-first approach is most appropriate for the proposed TriMinder application. Designing the system primarily for Android smartphones will ensure optimal accessibility and relevance to the majority of the target users.

Q3. Most Frequently Used Apps

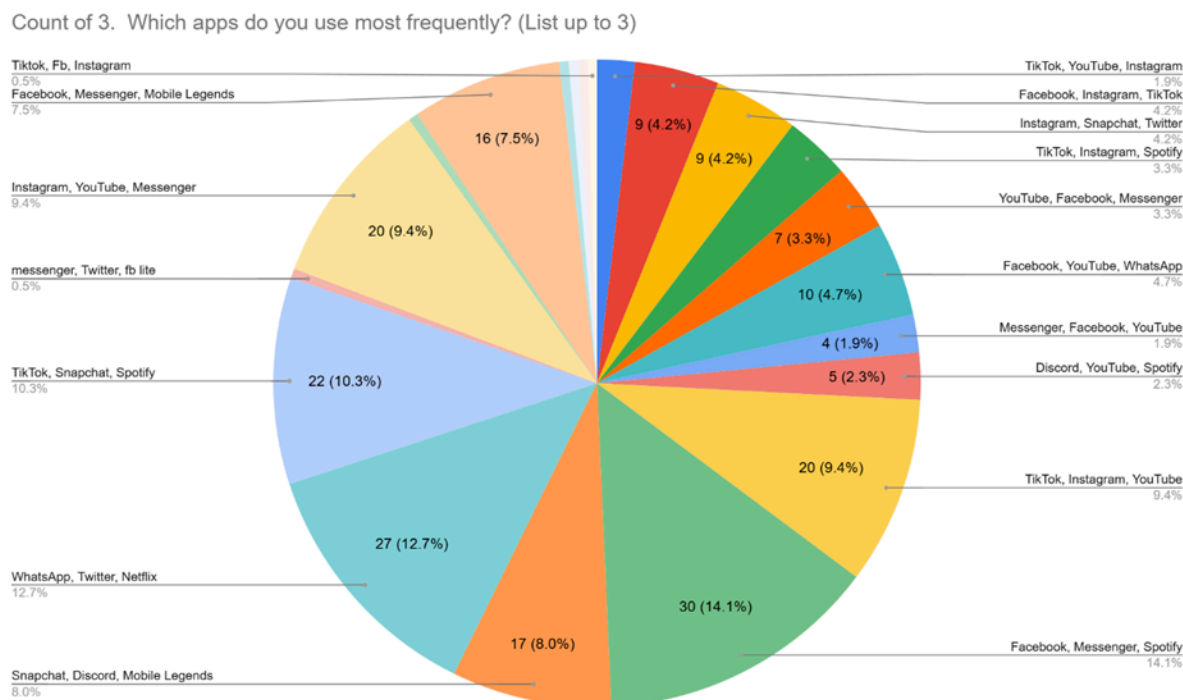


Figure 4.1.3 Most Frequently Used Apps

As shown in Figure 4.3, the top apps used by respondents include Facebook, TikTok, Messenger, YouTube, Instagram, and Spotify. These platforms are primarily for social interaction and entertainment, which may contribute to excessive screen time. TriMinder can address this trend by encouraging mindful usage through usage analytics and digital wellness rewards.

Q.4 Current Screen Time Tracking

4. Do you currently track or monitor your screen time usage?
211 responses

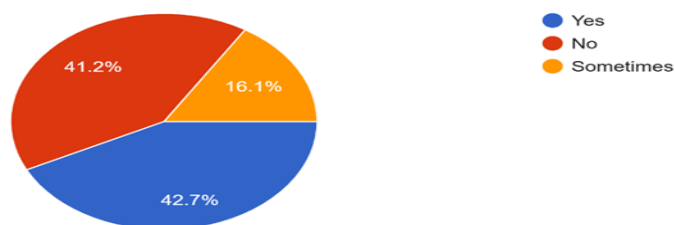


Figure 4.1.4 Current Screen Time Tracking

The results show that a large portion of students do not actively monitor their screen time, indicating low awareness of their digital habits. This gap suggests a strong opportunity for TriMinder to fill this role by providing user-friendly tracking features that raise awareness and encourage self-regulation.

Q5. Awareness of Screen Time Effects

5. Have you ever felt that excessive screen time affects your sleep, productivity, or physical health?
211 responses

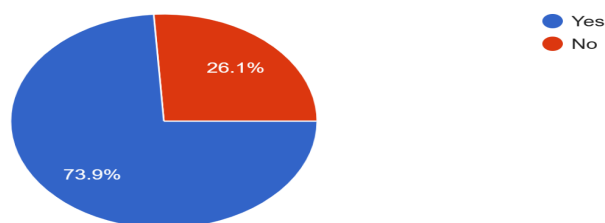


Figure 4.1.5 Awareness of Screen Time Effects

Figure 4.5 reveals that 73.9% of students acknowledge that excessive screen time negatively impacts their sleep, focus, or physical health. This data underscores the importance of

wellness education and reinforces the relevance of TriMinder's health-focused features, such as break notifications and sleep-time reminders.

Q6. Willingness to Use a Wellness App

6. Would you use an app that tracks your screen time and promotes wellness?
211 responses

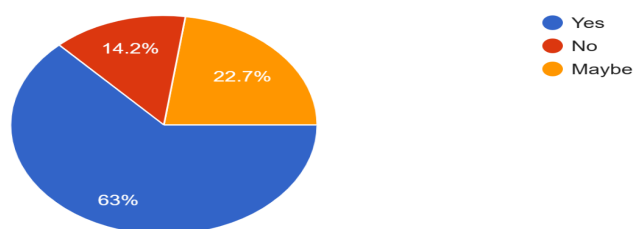


Figure 4.1.6 Willingness to Use a Wellness App

A majority of students expressed a positive response toward using an app that tracks screen time and promotes wellness. This validates TriMinder's overall concept and indicates strong user receptiveness to digital tools that support healthier technology use and lifestyle balance.

Q7. Importance of Break Reminders

7. How important is it for you to receive notifications reminding you to take breaks (e.g., stretch, rest eyes)?

211 responses

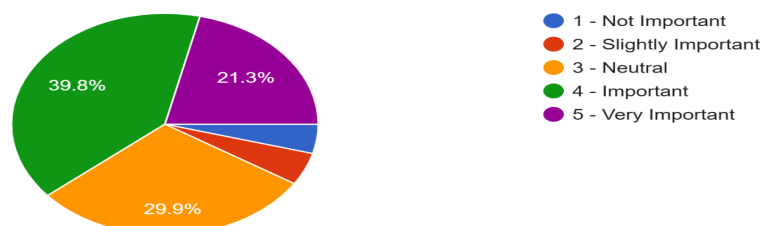


Figure 4.1.7 Importance of Break Reminders

According to the data, 91% of respondents find break notifications necessary, especially for activities like stretching or resting their eyes. This feature can serve as a key wellness component of TriMinder, helping prevent burnout and screen fatigue through timely and actionable prompts.

Q8. Motivation Through XP Points and Badges

8. Would features like XP points, badges, or streak rewards motivate you to reduce screen time?

211 responses

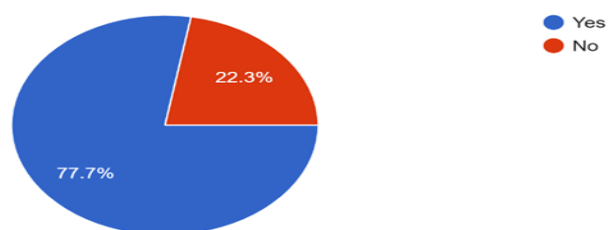


Figure 4.1.8 Motivation Through XP Points and Badges

The results show that 77.7% of students are motivated by gamification features such as XP points, badges, or streak rewards. This confirms that a gamified approach can be effective in promoting digital discipline, making wellness engagement more fun and sustainable within the TriMinder app.

Q9. Preferences for Personalization and Goal Setting

9. Do you prefer apps that allow personalization or goal setting (e.g., setting daily screen time limits)?
211 responses

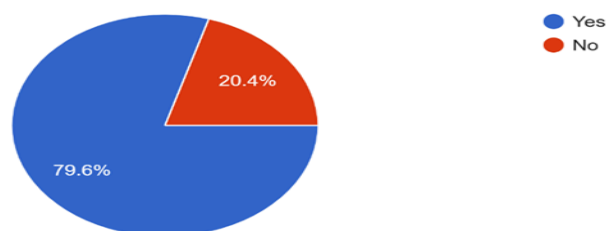


Figure 4.1.9 *Preferences for Personalization and Goal Setting*

A high 79.6% of respondents prefer apps that allow them to set personal goals or screen time limits. This indicates that customization is essential for user satisfaction, and TriMinder should include features that would enable students to tailor their experience to fit their unique habits and needs.

Q10. Motivation via Public Ranking

10. Would you be more motivated to manage your screen time if you were featured on a public ranking?

211 responses

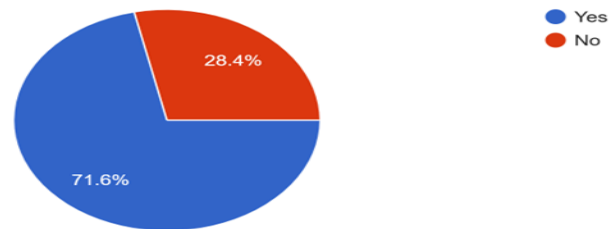


Figure 4.1.10 Motivation via Public Ranking

Figure 4.10 shows that 71.6% of students believe that being featured on a public ranking would motivate them to manage screen time better. This social visibility can drive friendly competition and accountability, making it a powerful feature for behavior change in the TriMinder app.

Q11. Viewing Friends' Progress

11. Would you like to see how your friends are performing in terms of digital discipline?

211 responses

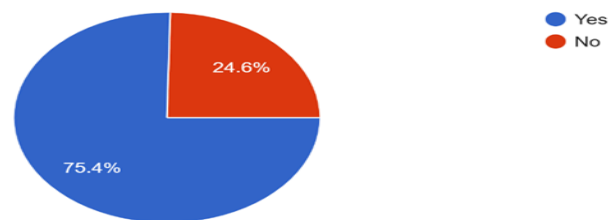


Figure 4.1.11 Viewing Friends' Progress

75.4% of students are interested in seeing their friends' progress in digital discipline, which supports the inclusion of community or peer interaction features in TriMinder. This can foster a sense of collective wellness and mutual encouragement among users.

Q12. Profile Like/Unlike Feature

12. Would the ability to "like" or "unlike" friends' performance profiles be meaningful for you?
211 responses

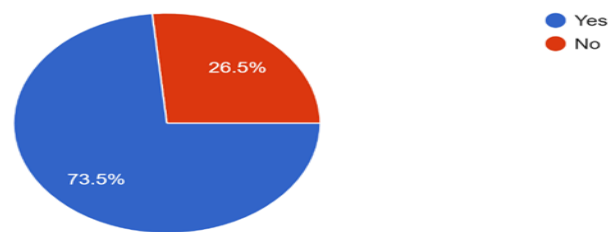


Figure 4.1.12 Profile Like/Unlike Feature

With 73.5% showing interest in the ability to 'like' or 'unlike' their friends' wellness profiles, this social feature could enhance user interaction and motivation. It introduces a simple way for students to acknowledge each other's progress, adding an extra layer of engagement in the app.

Q13. Trust in Student Leader for Recognition

13. Would you trust your student leaders (SSG or department orgs) to recognize top-performing students in a leaderboard?

211 responses

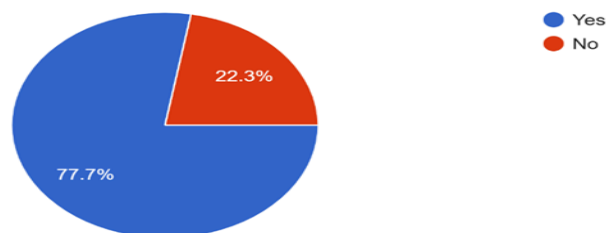


Figure 4.1.13 Trust in Student Leader for Recognition

A majority of students expressed trust in student leaders or organizations like the SSG to recognize top-performing users on a leaderboard. This shows a favorable attitude toward integrating wellness initiatives into student governance, adding credibility and incentive to TriMinder's leaderboard system.

Q14. Likelihood of Recommending the App

14. How likely are you to recommend a wellness-focused screen tracking app to your classmates?

211 responses

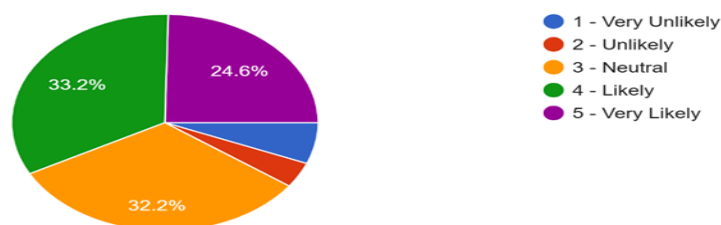


Figure 4.1.14 Likelihood of Recommending the App

Survey data shows that students are highly likely to recommend a wellness-focused screen tracking app like TriMinder to classmates, especially if it proves effective in promoting healthy habits. This reflects strong potential for organic app growth through peer endorsement and user satisfaction.

B. Thematic Analysis of Open-Ended Responses

Theme 1: Digital Overuse and Academic Strain

Many students described being unable to focus during study hours due to habitual app checking. The lack of control over their screen time leads to stress, cramming, and missed academic deadlines. Students frequently expressed frustration over procrastinating due to mindless scrolling, particularly at night or between classes.

Theme 2: Desire for Gamification

Students were notably receptive to gamified systems. They mentioned that achieving milestones, collecting XP, and receiving badges can positively reinforce better habits. Some suggested comparison features like “streaks” and “top 10 students per department” as added motivation.

Theme 3: Peer Influence and Social Comparison

The ability to see friends’ progress and get “likes” for good behavior was perceived as a healthy form of motivation. Several responses highlighted that such visibility builds accountability and that a public leaderboard can encourage friendly competition.

Theme 4: Customization and Usability

Students emphasized the need for features like class-synced break reminders, battery-saving UI, and a low-storage footprint. Others expressed concern about privacy and proposed anonymized rankings to reduce pressure while still fostering motivation.

C. Stakeholder Interviews (SSG and Department Organization)

In discussions with SSG officers and department student organization leaders, the need for administrative features became very clear. The SSG emphasized the value of a system that enables non-intrusive monitoring, supports report generation, and celebrates digitally disciplined students through recognition programs or digital awards.

Student organization officers emphasized the desire to view filtered rankings by department, which could be used during student assemblies or intramurals to recognize balanced digital habits. They also raised concerns over data privacy and preferred view-only dashboards with anonymized data when needed.

Stakeholders widely supported a gamified app that would reward students, enable inter-department competition, and provide admin dashboards for easy tracking.

Problem-Solution Fit

There is an apparent disconnect between current practices and desired digital behavior outcomes. While students are aware of their excessive screen usage, they lack the tools, community support, and motivation to change.

The proposed TriMinder mobile app and admin web platform directly address these needs through:

- Gamified tracking of screen time behavior.
- XP points and badge rewards for digital discipline.
- Daily rankings and peer interaction via profile likes.
- Admin dashboards for SSG and sub-admins to view reports and promote wellness.
- Offline data caching and cloud sync for hybrid access.

These solutions align with international standards for usability and performance, and integrate user feedback collected through mixed-method research.

4.2 Requirement Analysis

Functional Requirements

- FR1: The system shall monitor and record daily screen time for Android users.
- FR2: The system shall award XP points and badges based on healthy digital habits.
- FR3: The system shall allow students to add friends and view "Friends Ranking."
- FR4: The system shall enable users to like or unlike peer profiles once per day.
- FR5: The system shall provide daily department-based and general student rankings.
- FR6: The system shall generate break notifications (e.g., stretch, rest eyes).
- FR7: The system shall allow SSG super admins to manage department sub-admins
- FR8: The admin platform shall allow sub-admins to view department-only reports.
- FR9: The system shall display individual user profiles with wellness stats and badges.
- FR10: The system shall allow syncing with cloud storage and offline SQLite caching.
- FR-11: The system shall export user screen time reports (e.g., before graduation).

Non-Functional Requirements

- NFR1: The system shall respond to user actions within 2 seconds.
- NFR2: The system shall maintain usability for students with basic mobile experience.
- NFR3: The system shall ensure secure login and data encryption.
- NFR4: The system shall be compatible with Android 8 and above.
- NFR5: The system shall consume less than 100MB of storage on mobile devices.
- NFR6: The system shall support dark mode and energy-efficient UI.
- NFR7: The system shall work offline and sync data when an internet connection is available.
- NFR8: The system shall be maintainable through modular code using Flutter and Supabase.

User Requirements

- Student Users: Can track screen usage, earn XP, add friends, like profiles, and view rankings.
- Sub-admin (Dept Org Officer): Can view department-specific rankings, encourage users via likes.
- Super Admin (SSG Officer): Can manage sub-admin access, view all student rankings, and recognize top users.

Table 4.2.1 System Requirements Table

Requirement ID	Type	Description	Priority
FR1	Functional	The system shall monitor and record daily screen time for Android users.	High
FR2	Functional	The system shall award XP points and badges based on healthy digital habits.	High
FR3	Functional	The system shall allow students to add friends and view "Friends Ranking.	Medium
FR4	Functional	The system shall enable users to like or unlike peer profiles once per day.	High
FR5	Functional	The system shall provide daily department-based and general student rankings.	High
FR6	Functional	The system shall generate break notifications (e.g., stretch, rest eyes).	Medium
FR7	Functional	The system shall allow SSG super admins to manage department sub-admins.	High
FR8	Functional	The admin platform shall allow sub-admins to view department-only reports.	Medium
FR9	Functional	The system shall display individual user profiles with wellness stats and badges.	Medium
FR10	Functional	The system shall allow syncing with cloud storage and offline SQLite caching.	High
FR11	Functional	The system shall export user screen time reports (e.g., before graduation).	Low

NFR1	Non-Functional	The system shall respond to user actions within 2 seconds.	High
NFR2	Non-Functional	The system shall maintain usability for students with basic mobile experience.	High
NFR3	Non-Functional	The system shall ensure secure login and data encryption.	Critical
NFR4	Non-Functional	The system shall be compatible with Android 8 and above.	High
NFR5	Non-Functional	The system shall consume less than 100MB of storage on mobile devices.	Medium
NFR6	Non-Functional	The system shall support dark mode and energy-efficient UI.	Medium
NFR7	Non-Functional	The system shall work offline and sync data when online.	High
NFR8	Non-Functional	The system shall be maintainable through modular code using Flutter and Supabase.	Medium
UR1	User	Students can track usage, earn XP, add friends, like profiles, and view rankings.	High
UR2	User	Sub-admins can view department-specific rankings and send likes.	Medium
UR3	User	Super admins can manage sub-admin access and recognize top users.	High

CHAPTER 5: PROJECT PLAN

5.1 Project Schedule

The development of TriMinder follows the **Agile Software Development Life Cycle (SDLC)** methodology, which allows for iterative development with continuous user feedback and progressive system refinement. The project is structured across **18 weeks** to align with the academic calendar and ensure thorough development, testing, and documentation phases.

Project Phases and Activities

The TriMinder project is divided into seven major phases, each containing specific activities and deliverables:

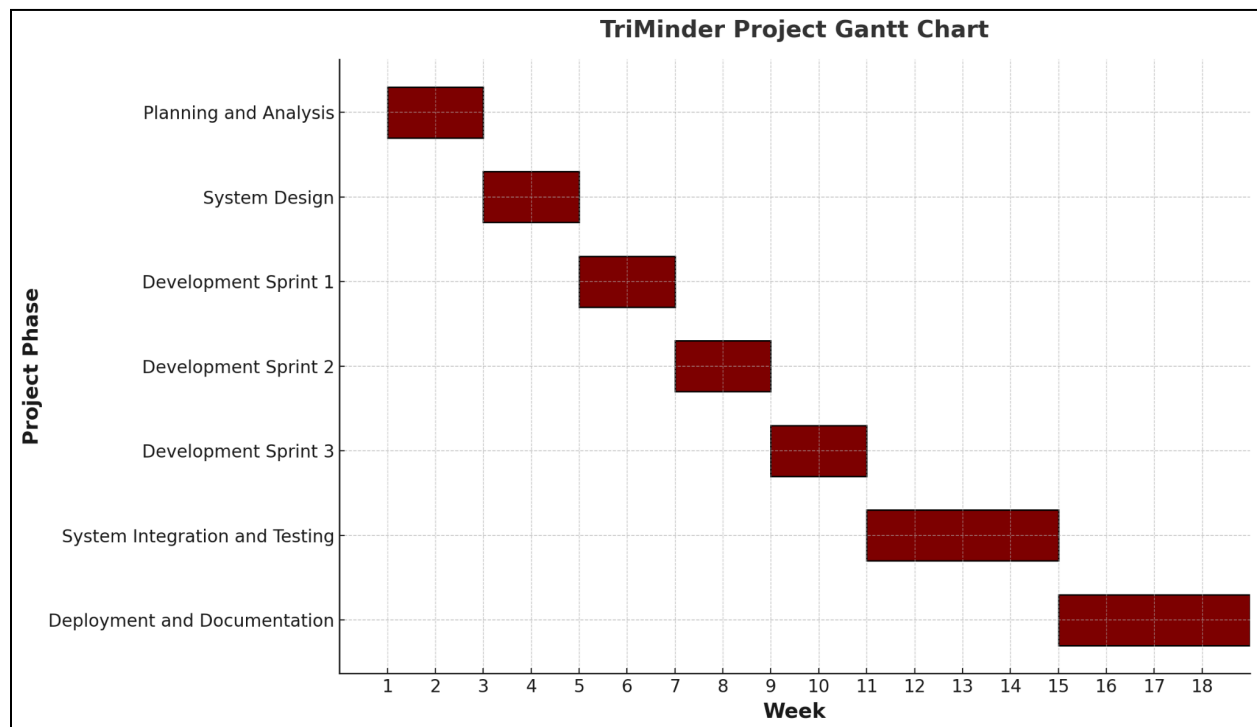


Figure 5.1.1 Proposed Project Timeline for TriMinder (Week 1 to Week 18)

Phase 1: Planning and Analysis (Weeks 1-2)

- Conduct comprehensive stakeholder surveys and interviews
- Perform literature synthesis and review existing solutions
- Finalize functional and non-functional requirements
- Establish project scope and limitations
- Complete needs assessment and requirement analysis

Phase 2: System Design (Weeks 3-4)

- Create system architecture and database design
- Develop Entity-Relationship Diagrams (ERD) and Data Flow Diagrams (DFD)
- Design wireframes and user interface mockups using Figma
- Establish UI/UX style guide and design standards
- Build low-fidelity prototypes for user validation

Phase 3: Development Sprint 1 (Weeks 5-6)

- Implement core screen time monitoring functionality
- Develop a user authentication system using Supabase
- Create a local database schema using SQLite
- Build fundamental tracking modules
- Conduct unit testing for core components

Phase 4: Development Sprint 2 (Weeks 7-8)

- Implement gamification engine (XP points, badges, streaks)
- Develop reward and achievement systems
- Create user dashboard and progress tracking interfaces

- Integrate a notification system for wellness breaks
- Conduct a sprint demo with pilot users

Phase 5: Development Sprint 3 (Weeks 9-10)

- Develop social features (friend system, ranking, likes)
- Implement an admin web platform for SSG and sub-admins
- Create department-based filtering and reporting
- Add real-time synchronization between mobile and web platforms
- Integrate all system modules and conduct integration testing

Phase 6: System Integration and Testing (Weeks 11-14)

- Perform comprehensive system integration
- Conduct functional, security, and performance testing
- Execute usability testing based on ISO/IEC 25010 standards
- Gather user feedback and iterate on UI/UX improvements
- Address identified defects and optimization issues

Phase 7: Deployment and Documentation (Weeks 15-18)

- Prepare application for Android deployment
- Complete comprehensive project documentation
- Conduct final system evaluation and metrics compilation
- Prepare for oral defense and panel presentation
- Execute project turnover and maintenance planning

Duration and Dependencies

- Critical dependencies have been identified to ensure smooth project progression:
- Sprint 2 cannot commence until Sprint 1 delivers stable API foundations and user authentication systems
- Integration Testing (Weeks 11-12) requires completion of all three development sprints
- Usability Testing (Weeks 13-14) depends on a stable beta build from the integration phase
- Deployment Preparation (Week 15) requires finalized UI/UX elements and branding assets from the Design phase
- Final Documentation cannot be completed until all testing phases provide comprehensive evaluation results

5.2 Milestones and Deliverables

The TriMinder project defines **nine major milestones** that serve as critical checkpoints for project progress evaluation and quality assurance. Each milestone produces specific, measurable deliverables that contribute to the overall project success.

Table 5.2.1 Milestone Table

Milestone No.	Description	Key Deliverable	Target Week	Success Criteria
M1	Requirements Analysis Completion	Requirements Specification Document	Week 2	Approved functional and non-functional requirements with stakeholder sign-off

M2	System Design Approval	ERD, DFD, and Wireframes Package	Week 4	Complete system architecture with adviser approval
M3	Core Functionality Demo	Functional Prototype (Tracking + Authentication)	Week 6	Working screen time monitoring with user login capabilities
M4	Gamification System Demo	Integrated Prototype with Rewards System	Week 8	Functional XP points, badges, and achievement tracking
M5	Social Features Integration	Complete Mobile App with Admin Web Platform	Week 10	Working friend system, ranking, and admin dashboard
M6	System Integration Completion	Integrated Beta Release	Week 12	Fully integrated system with zero Priority 1 bugs
M7	Usability Evaluation Sign-off	ISO/IEC 25010 Usability Report	Week 14	System Usability Scale (SUS) score ≥ 80 with positive user feedback
M8	Deployment Readiness	Release Candidate APK and Web Platform	Week 15	Production-ready applications with security validation
M9	Project Completion & Oral Defense	Final Manuscript, User Manual, Signed Turnover	Week 18	Approved final manuscript with successful oral defense

Risk Mitigation and Contingency Planning

Each milestone incorporates buffer time and alternative approaches to address potential delays:

- **Technical Risks:** Backup development approaches identified for complex integrations
- **Resource Risks:** Cross-training among team members to ensure continuity
- **Schedule Risks:** Priority-based feature implementation allowing scope adjustment if necessary
- **Quality Risks:** Continuous testing and integration throughout development phases

CHAPTER 6: TECHNICAL SPECIFICATION

6.1 Hardware Requirements

The TriMinder application requires specific hardware for development and user operation. The following specifications ensure optimal performance across all system components.

Table 6.1.1 Development Hardware

Component	Minimum Specification	Recommended Specification	Purpose
Developer Computer	Intel Core i5, 8GB RAM, 256GB SSD	Intel Core i7, 16GB RAM, 512GB SSD	Development and testing
Android Phone	Android 7.0+, 3GB RAM, 32GB storage	Android 10.0+, 6GB RAM, 64GB storage	App testing
Internet Connection	5 Mbps broadband	25 Mbps broadband	Cloud services access

6.2 Software Requirements

TriMinder's development leverages open-source tools and industry-standard frameworks to ensure scalability, maintainability, and cross-platform compatibility. The following tools and technologies are required throughout the application lifecycle.

Table 6.2.1 Development Software

Software/Tool	Version	Function
Flutter SDK	V3.24.0 or later	Mobile app development framework

Android Studio	Latest Version	Development Environment
Visual Studio Code	Latest Version	Code Editor (Alternative)
Git	Latest Version	Version Control
Android SDK	API Level 24-34	Android Development Tools

Table 6.2.2 Backend Services

Service	Type	Function
Supabase	Cloud-hosted	Database and authentication
PostgreSQL	Database (via Supabase)	Cloud data storage
SQLite	Local Database	Offline data storage and caching
Supabase Auth	Authentication service	User login and security
Supabase Storage	File storage	User data and app assets

Table 6.2.3 Design Tools

Tool	Type	Function
Figma	Web-based	UI/UX design
Draw.io	Web-based	System diagrams
Lucidchart	Web-based	System architecture and flowchart diagrams
GitHub	Cloud service	Code repository

Table 6.2.4 Target Platform

Platform Component	Requirement	Purpose
Android OS	Version 7.0 (API 24) minimum	App runtime environment
SQLite	Version 3.31+	Local database operations
Internet Browser	Chrome 70+ or equivalent	In-app web content

6.3 Network Requirements

TriMinder requires internet connectivity for synchronization, authentication, and cloud features. The application also supports offline functionality for core features through SQLite local storage.

Table 6.3.1 Internet Connectivity

Requirement	Specification	Purpose
Minimum Speed	1 Mbps download, 512 Kbps upload	Basic app functionality
Recommended Speed	5 Mbps download, 2 Mbps upload	Optimal performance
Connection Type	Wi-Fi or mobile data (4G/5G)	Flexible connectivity
Data Usage	10-50 MB per month	Estimated monthly consumption

Table 6.3.2 Network Features

Feature	Specification	Benefit
Offline Mode	Core features work without internet	Uninterrupted usage
Auto-Sync	Automatic data synchronization when online	Data consistency between local and cloud
Security	HTTPS/TLS encryption for all data	Secure communication
Cloud Backup	Automatic backup to Supabase Pro	Data protection and recovery

Table 6.3.3 Development Network

Requirement	Specification	Purpose
Development Internet	25 Mbps minimum	Code repository access
Testing Connection	Stable broadband	App testing and debugging
Deployment Access	High-speed upload	App store publishing

6.4 Chapter Summary

The TriMinder application is designed to run efficiently on modern Android devices, requiring minimal hardware resources. The development environment uses industry-standard tools, including Flutter for cross-platform development, Supabase Pro tier for backend services, and SQLite for local data storage and offline functionality.

The dual-database approach ensures the app remains functional even with limited connectivity through SQLite local storage, while maintaining cloud synchronization capabilities through the Supabase Pro tier. Network requirements are optimized for both online and offline usage, ensuring robust performance across various connectivity scenarios.

The technical specifications support the project's goal of creating an accessible, gamified wellness application that can operate effectively across a wide range of Android devices while maintaining security, performance, and offline capability standards.

CHAPTER 7: BUDGET AND RESOURCES

7.1 Cost Breakdown

Summary of Direct vs Indirect Costs

- Direct costs are expenses explicitly tied to designing, developing, testing, and releasing TriMinder.
- Indirect costs support general academic activities (e.g., snacks) or provide contingency buffers.

Table 7.1.1 Itemized Budget Table

Classification	Item / Description	Unit Cost	Quantity	Sub-Total	Funding Source
DIRECT	Document printing (draft & final, A4)	₱5 /page	100 pages	₱500	Proponents
	Flash drive 32 GB (project backup)	₱270 /pc	1	₱270	Proponents
	Smart/TNT prepaid load cards (data for mobile testing)	₱100 /ea	10	₱1,000	Proponents
	Supabase “Pro” plan (1 month) for production DB	\$25 ≈ ₱1,400	1 month	₱1,400	Proponents
	Contingency (10% of direct total)	—	—	₱317	Proponents
DIRECT SUB-TOTAL	—	—	—	₱3,487	—

INDIRECT	Meals & snacks for 30-participant usability test	₱100 per participant	30	₱3,000	Proponents
	Misc. office supplies (bond paper, ink, sticky notes)	₱500 lump-sum	1	₱500	Proponents
INDIRECT SUB-TOTAL	—	—	—	₱3,500	—
GRAND TOTAL	—	—	—	₱6,987	—

Exchange rate used: US\$1 = ₱57 (Aug 2025).

7.2 Materials and Logistics

Table 7.2.1 Manpower and Roles

Role	Primary Tasks	Assigned Member	Outside Expertise Needed
Project Manager / Back-End Dev	Sprint planning, Supabase schema, API security	Jet Venson G. Romero	Supabase community forum
Mobile App Lead	Flutter UI, Android integration, unit testing	Russel Jhon C. Dasigan	None
UI/UX & QA	Wireframes, ISO/IEC 25010 usability testing, bug triage	Maria Angie C. Niez	Adviser feedback for heuristic evaluation
Capstone Adviser	Technical oversight, SDLC milestone approval	TBA	—
Volunteer Testers (30)	Usability & pilot study participants	Recruited students	—

Table 7.2.2 Equipment and Tools Inventory

Resource Type	Description	Availability	Remarks
Development laptops	Intel i5/8 GB RAM or higher (3 units)	Available	Personal devices of proponents
Android test phones	Android 10+, 4 GB RAM (3 units)	Available	Ensures wide OS coverage
High-speed internet	25 Mbps fiber at campus lab	Available	Primary coding & video-call channel
Mobile data packages	10 × Smart load ₱100	Procured	Real-world connectivity tests
Supabase backend	Pro tier	Provisioned	Scales for pilot users
GitHub private repo	Student discount (free)	Activated	Source control & CI
Design software	Figma (education plan)	Free	Interface prototypes & assets
Printing services	Local print shop offering ₱5/page	Accessible	Quick turnaround, Ormoc CBD
Binding service	Softbind ₱260/book average	Available	For final manuscript submission

Support Services and Facilities

- **Computer Laboratory (EVSU–Ormoc):** Fallback build environment and group sprint reviews.
- **Campus Wi-Fi Hotspots:** Secondary internet channel for remote collaboration.
- **Local Print Shop:** Document printing and binding services for manuscripts and presentations.

Logistical Plan

- **Usability Testing:** Week 13 in campus lab (08:00–17:00); two 2-hour sessions with snacks.
- **Printing & Binding:** Week 17 at local print shop; prepare draft and final copies.
- **Final Deployment:** Week 15; Supabase Pro plan active and APK tested on target devices.

Resource Sufficiency Statement

Proponents' devices and free academic licenses cover all development needs. The allocated budget ensures at least one month of production hosting post-defense, allowing for final bug fixes and minor enhancements. Available equipment, workforce, and facilities confirm that TriMinder is both financially and logistically feasible within the academic term.

CHAPTER 8: CONCLUSION

This proposal addresses the critical problem of excessive and unregulated screen time among college students at Eastern Visayas State University (EVSU) Ormoc Campus, which negatively impacts their academic focus, sleep routines, physical activity, and mental wellness. The proposed system, titled "TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits," is designed to provide a reliable and user-friendly solution to improve digital discipline, wellness awareness, and peer-driven motivation through gamified engagement strategies.

The general objective of this project is to develop a comprehensive gamified screen time management and wellness monitoring application that offers interactive tracking, social engagement, and administrative oversight capabilities. Specifically, the system will provide core functionalities including real-time screen usage monitoring, gamification features such as XP points and digital discipline badges, friend-based ranking systems with peer interaction capabilities, wellness break notifications, and comprehensive administrative platforms for both SSG super admins and department-based sub-admins. These components are carefully aligned with the data gathered during the needs assessment phase, where over 200 EVSU Ormoc students expressed strong interest in gamified wellness solutions and peer-to-peer engagement features.

With the successful implementation of this project, the primary beneficiaries, such as EVSU Ormoc college students, are expected to gain improved digital self-discipline, enhanced awareness of screen usage patterns, reduced technology-related strain, and increased motivation through peer recognition and healthy competition. The gamified approach, incorporating XP points, badges, and daily rankings, will foster sustained engagement, while the friend system and profile interaction features will create a supportive community environment. Additionally, this system contributes to ongoing efforts toward digital transformation in higher education, aligning with institutional modernization goals and the Commission on Higher Education's advocacy for digital wellness and productivity in higher education institutions.

The feasibility of the project has been confirmed through careful planning, realistic cost estimation totaling ₱6,987, availability of appropriate development tools including Flutter SDK, Supabase Pro backend services, and SQLite for offline functionality, and the technical competency of the development team. The Agile Software Development Life Cycle methodology ensures iterative development with continuous user feedback integration across 18 weeks of structured development phases. All necessary resources, including development hardware, software licenses, network infrastructure, and stakeholder support mechanisms, have been identified and secured through a combination of personal resources and institutional facilities.

Following the approval of this proposal, the team will proceed to the comprehensive development phase, structured across seven distinct phases: Planning and Analysis, System Design, three iterative Development Sprints, System Integration and Testing, and finally Deployment and Documentation. This will include complete system architecture implementation, extensive usability testing based on ISO/IEC 25010 standards, user feedback integration through pilot deployments, and thorough quality assurance validation. The project team is committed to delivering a high-quality, gamified wellness solution that meets user needs, promotes sustainable digital habits, and supports the broader institutional goals of student wellness and academic excellence through innovative technology integration.

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APPENDICES

Appendix A – Survey Questionnaire

Survey Questionnaire: TriMinder User Insight Form

Title of the Study:

TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits among EVSU Ormoc Students

Purpose of the Study:

This survey aims to gather insights from EVSU Ormoc students regarding their screen time habits, preferences, and openness to using a gamified wellness tracking application. The data collected will help in designing *TriMinder*, a mobile application that promotes healthy digital habits through reminders, rewards, and peer interaction. Your honest responses will contribute to developing a system tailored to the needs and behaviors of students like you.

Instructions:

Please answer the following questions honestly. Your responses will remain confidential and will be used solely for research purposes.

Personal information

- Name: _____
- Age: _____
- Year Level: _____

- Section: _____
- Department: _____

Screen Time & Wellness Questions

- On average, how many hours a day do you use your smartphone?
 ____ hours
- What type of device do you primarily use?
☐ Android ☐ iOS (Apple) ☐ Other: _____
- Which apps do you use most frequently?
 List up to 3: _____
- Do you currently track or monitor your screen time usage?
☐ Yes ☐ No
- Have you ever felt that excessive screen time affects your sleep, productivity, or physical health?
☐ Yes ☐ No
- Would you use an app that tracks your screen time and promotes wellness?
☐ Yes ☐ No
- How important is it for you to receive notifications reminding you to take breaks (e.g., stretch, rest eyes)?
☐ 1 – Not important ☐ 2 ☐ 3 – Neutral ☐ 4 ☐ 5 – Very important

7. Would features like XP points, badges, or streak rewards motivate you to reduce screen time?
- ☐ Yes ☐ No
8. Do you prefer apps that allow personalization or goal setting (e.g., setting daily screen time limits)?
- ☐ Yes ☐ No
9. Would you be more motivated to manage your screen time if you were featured on a public ranking?
- ☐ Yes ☐ No
10. Would you like to see how your friends are performing in terms of digital discipline?
- ☐ Yes ☐ No
11. Would the ability to “like” or “unlike” friends’ performance profiles be meaningful for you?
- ☐ Yes ☐ No
12. Do you think friendly competition can help students be more mindful of their screen usage?
- ☐ Yes ☐ No
13. How likely are you to recommend a wellness-focused screen-tracking app to your classmates?
- ☐ 1 – Very unlikely ☐ 2 ☐ 3 – Neutral ☐ 4 ☐ 5 – Very likely

14. Any additional comments, feedback, or suggestions?

Consent Declaration

☐ I confirm that I have read the purpose of this survey and voluntarily agree to participate. I understand that all data will be treated confidentially and in an anonymous manner.

Signature (or initials): _____

Date: _____

Appendix B – System Design Artifacts

B.1 Entity-Relationship Diagram (ERD)

This diagram represents the database structure for the TriMinder application, showing the relationships between users, screen time data, gamification elements, and guardian controls.

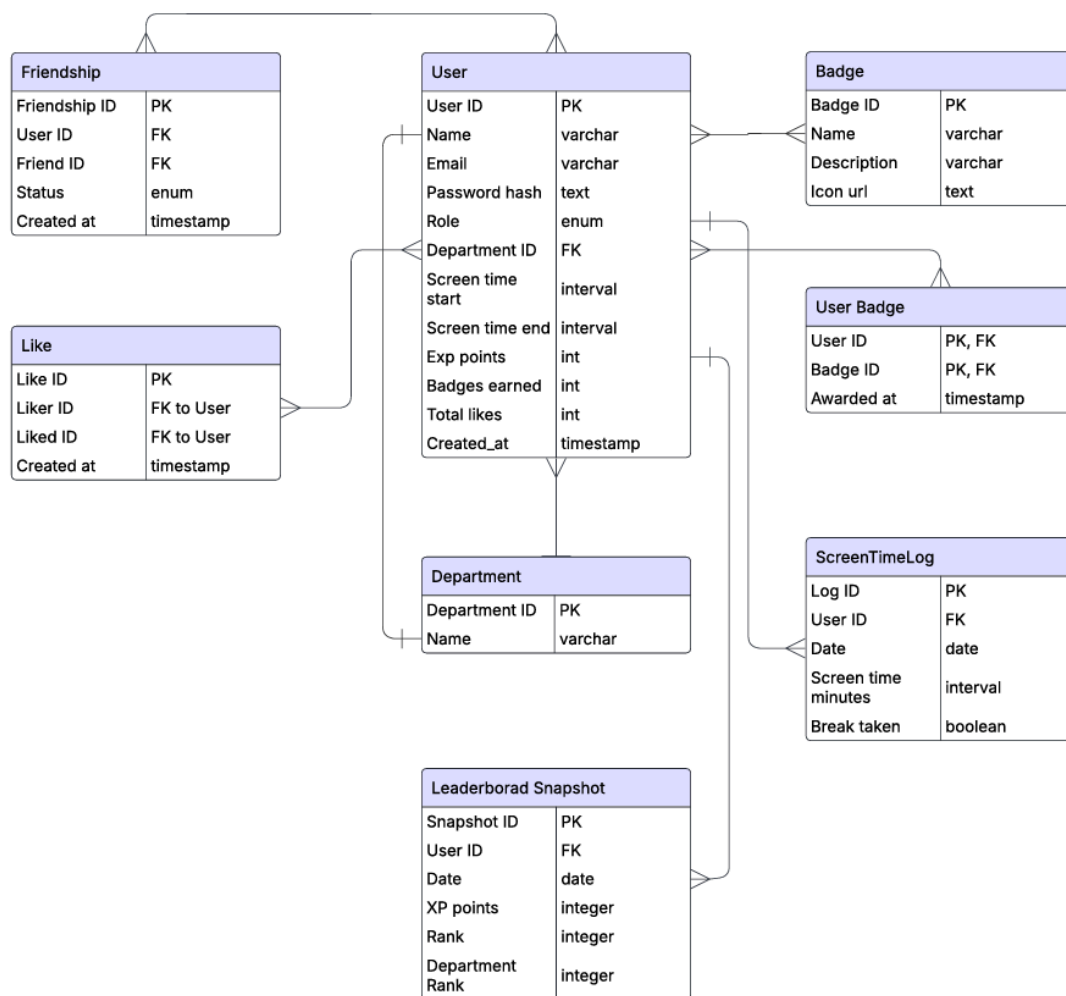


Figure B.1 Entity-Relationship Diagram (ERD) for TriMinder

Appendix C – Informed Consent Forms

C.1 Consent Form – Student Respondents

Informed Consent Form – Student Participants

Project Title:

TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits among EVSU Ormoc Students

Dear Participant,

You are invited to participate in a survey as part of our capstone research. The goal of this project is to develop a wellness and screen time monitoring app designed for students like you. The survey will take approximately 5–10 minutes.

Your participation is voluntary, and your identity will remain anonymous. Your answers will only be used for academic purposes and will help us improve the effectiveness of the app.

By participating, you agree to the following:

- I understand that I can withdraw at any time.
- I understand that all data will be kept confidential.
- I voluntarily agree to participate in the survey.

Signed: _____

Date: _____

C.2 Consent Form – Student Organization Officer (Sub Admin)

Informed Consent Form – Department-Based Organization Officer

Dear Student Organization Officer,

We are conducting a capstone research project entitled *TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits Among EVSU Ormoc Students*. As a department-based officer, your insights will guide the sub-admin functionalities of our web system.

Your participation in this interview/survey is voluntary and will remain confidential. Your responses will help us understand how department organizations can promote digital wellness.

Please sign below to indicate your informed consent:

Name: _____

Department: _____

Signature: _____

Date: _____

C.3 Consent Form: SSG Representative (Super Admin)

Informed Consent Form – Supreme Student Government Representative

Dear SSG Representative,

You are invited to participate in our capstone research entitled *TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits Among EVSU Ormoc Students*. Your role as a super admin is critical to designing the administrative features of the system.

We request your input through an interview or questionnaire. Your participation is voluntary, and all responses will be kept confidential.

Please indicate your consent by signing below:

Name: _____

Position: _____

Signature: _____

Date: _____

Appendix D – Gantt Chart

The TriMinder project follows a structured timeline encompassing seven key phases: Planning and Analysis (Weeks 1-2), System Design (Weeks 3-4), three sequential Development Sprints (Weeks 5-10), System Integration and Testing (Weeks 11-14), and Deployment and Documentation (Weeks 15-18). This phased approach ensures systematic progress from initial analysis through development and thorough testing to final deployment, optimizing project management and delivery within an 18-week timeframe.

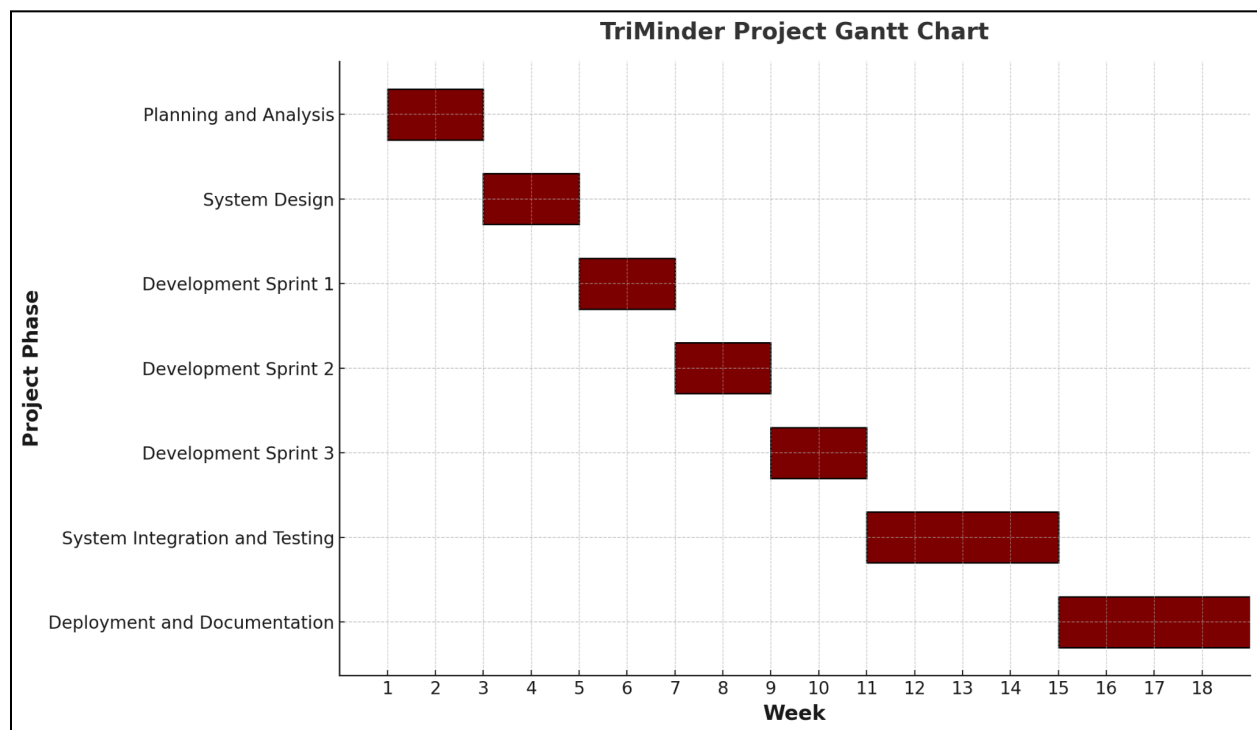


Figure D.1 Gantt Chart for TriMinder Project Timeline

Appendix E – Context-Level Data Flow Diagram (DFD) for TriMinder

This appendix presents the Context-Level Data Flow Diagram (DFD) for the TriMinder system. This facilitates user authentication, social interactions, and administrative functions, integrating features like screen time tracking, notifications, and gamification management. It manages data flow among users, departments, and system modules to provide a comprehensive notification and engagement platform.

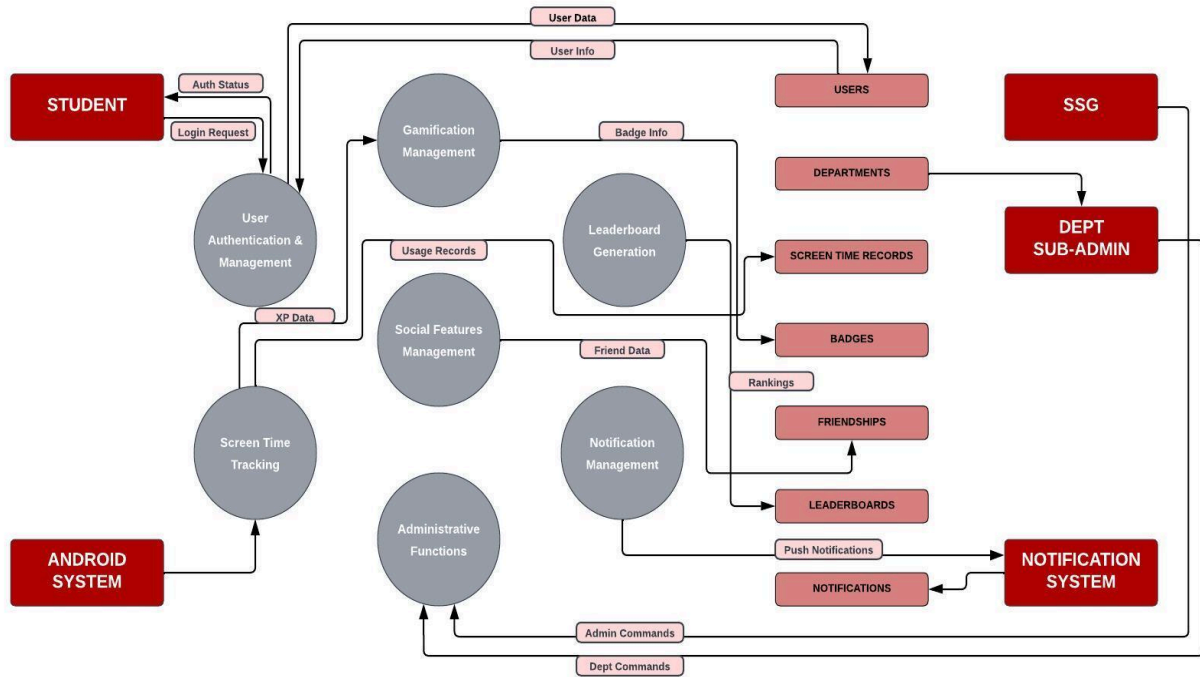


Figure E.1 Context-Level Data Flow Diagram (DFD) for TriMinder

